Arista Access Node QoS/Congestion Event Troubleshooting and Resolution Standard Operating Procedure (SOP)

Node: KR-ACC-82 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_1245

Description

The QoS/Congestion Event failure domain, denoted by alarm code ALARM_1245, occurs on Arista access nodes due to microbursts, interface tail drops, or QoS queue starvation. This phenomenon arises from the intricate interplay between packet scheduling, queue management, and traffic policing mechanisms within the node's control and data planes. Specifically, on Arista equipment, the EOS (Extensible Operating System) implements a hierarchical QoS framework, comprising class-maps, policy-maps, and service-policies, which dictate traffic classification, marking, and policing. When the node experiences a surge in traffic, potentially exacerbated by improper QoS configuration or inadequate bandwidth allocation, it may lead to queue overflows, resulting in packet drops and performance degradation. Furthermore, the Arista's buffered output mechanism, designed to mitigate the impact of microbursts, may itself become overwhelmed, precipitating a cascade of congestion-related issues. This SOP aims to provide a structured approach to diagnosing and resolving such QoS/Congestion Events on Arista access nodes, ensuring prompt restoration of optimal network performance.

Pre-requisite Checks

- Verify redundant link status to ensure no single point of failure
- Dispatch remote hands to KR Puram Access for potential on-site interventions
- Confirm all relevant network monitoring tools are operational and providing real-time data

Potential Root Causes

Cause	Probability
Misconfigured QoS policies or class-maps	High
Inadequate interface bandwidth or improper traffic policing	Medium
Hardware or firmware issues affecting queue management	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos interface ethernet 1, Expected Output: Displays the current QoS configuration and statistics for the specified interface, Description: This command isolates the QoS configuration applied to the interface, including class-maps, policy-maps, and service-policies, allowing for the identification of potential misconfigurations or mismatches between configured and operational
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parameters.

Command: show buffers, Expected Output: Displays buffer utilization
and allocation statistics, Description: By examining buffer
utilization, this command helps identify if the node's buffered output
mechanism is being overwhelmed, potentially indicating
microburst-related issues or insufficient buffer allocation.
```

Resolution Steps

Step 1: Step: Adjust QoS Policies and Class-Maps, Command: configure terminal ; interface ethernet 1 ; service-policy input my-qos-policy, Description: Modify the QoS policies and class-maps to ensure proper traffic classification, marking, and policing, preventing queue overflows and performance degradation.

Step 2: Step: Optimize Interface Bandwidth and Traffic Policing, Command: configure terminal ; interface ethernet 1 ; bandwidth 10000 ; police rate 5000 burst 1000, Description: Adjust the interface bandwidth and traffic policing parameters to match the actual traffic requirements, preventing congestion and ensuring that critical traffic is prioritized.

Post-Check Verification

```
Command: show qos interface ethernet 1, Expected Output: Displays the
updated QoS configuration and statistics, reflecting the changes made,
Description: Verify that the QoS policies and class-maps have been
correctly applied and are functioning as intended, with no signs of
queue overflows or performance degradation.

Command: show interfaces ethernet 1 statistics, Expected Output:
Displays interface statistics, including packet counts, errors, and
drops, Description: Confirm that the interface is operating within
expected parameters, with no unusual error rates or packet drops,
indicating successful resolution of the QoS/Congestion Event.
```

Roll-back Procedures

- Step: Revert QoS Configuration, Command: configure terminal ; no service-policy input my-qos-policy, Description: If the applied changes exacerbate the issue or do not resolve it, revert the QoS configuration to its previous state to prevent further complications.

- Step: Restore Original Interface Configuration, Command: configure terminal ; default interface ethernet 1, Description: If adjustments to the interface bandwidth or traffic policing worsen the situation, restore the interface to its default configuration to ensure a known baseline state.